FOR ARCHITECTS · SRE LEADS · AI PROGRAM OWNERS

ELASTICSEARCH + VECTOR DB + ELK + REDIS? ONE BINARY.

A head-to-head architecture brief for teams running — or about to run — the legacy retrieval stack under AI workloads. Vector, full-text, histograms, and agent memory through one engine, one wire protocol, and one bill. The voices in this brief are real, public, and dated.

RELEASE V1.0.0-RC.1 · ES 8.13 WIRE-COMPATIBLE · SELF-HOSTED · AGENT-READY

02 · AI WORKLOADS BREAK THE OLD STACK

AGENTS DON'T REPLACE THE RETRIEVAL STACK. THEY AMPLIFY ITS CRACKS.

Elasticsearch and the wider ELK stack carried a decade of search, logging, and SIEM workloads. They were not designed for the retrieval pattern an LLM agent imposes — many small, latency-bound, cross-modality calls per turn. Vendor-grade hybrid search, vector indexes, and agent memory have been bolted on. The architectural consequences land on the team running the cluster.

JVM HEAP

31 GB CEILING. GC PAUSES.

Above 31 GB heap, compressed object pointers turn off and usable memory halves. Below that, GC pauses and circuit breakers govern tail latency. Vector graphs want everything in RAM.

QUERY SHAPE

HYBRID RETRIEVAL IS A HAIR-BALL.

kNN + filter + BM25 + agg in one request requires post-filter workarounds, doubled scores, and rank-fusion in app code or a paid ranker. Score normalisation is the architect's problem.

SCALE TAX

RECOVERY STORMS. FIELD EXPLOSIONS.

Tens of thousands of small shards each consume a fixed slice of heap. A node restart triggers a recovery storm. Dynamic mappings turn ingest mistakes into cluster-state bloat.

VENDOR FAN-OUT

FOUR STORES. FOUR BILLS.

Most AI-native shops now run ES + a vector DB + Redis + something for logs. Four wire protocols, four quotas, four on-calls. The agent loop is one turn. The tail is four tails added together.

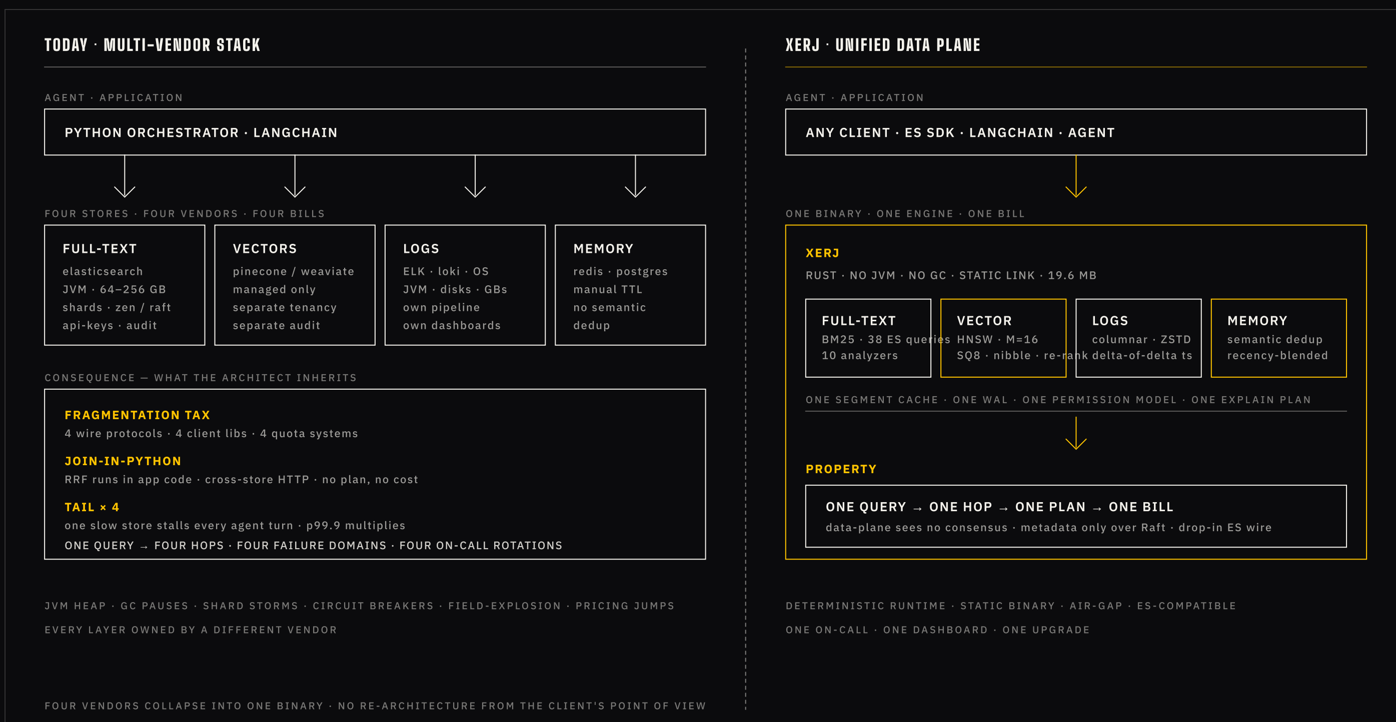
WHAT THIS BRIEF DOES

Walks an architect through the head-to-head: where the legacy stack hurts under AI workloads, where Xerj is structurally different, and what the published, real-user voices actually say. Quotes are dated. Sources are listed.

03 · THE STACK CHOICE

FOUR VENDORS.
OR ONE BINARY.

Today's AI shop runs the retrieval stack as a federation of point products, glued together in Python. Xerj collapses that federation into a single Rust binary that already speaks the Elasticsearch wire protocol. The agent code does not change. The architecture diagram does.



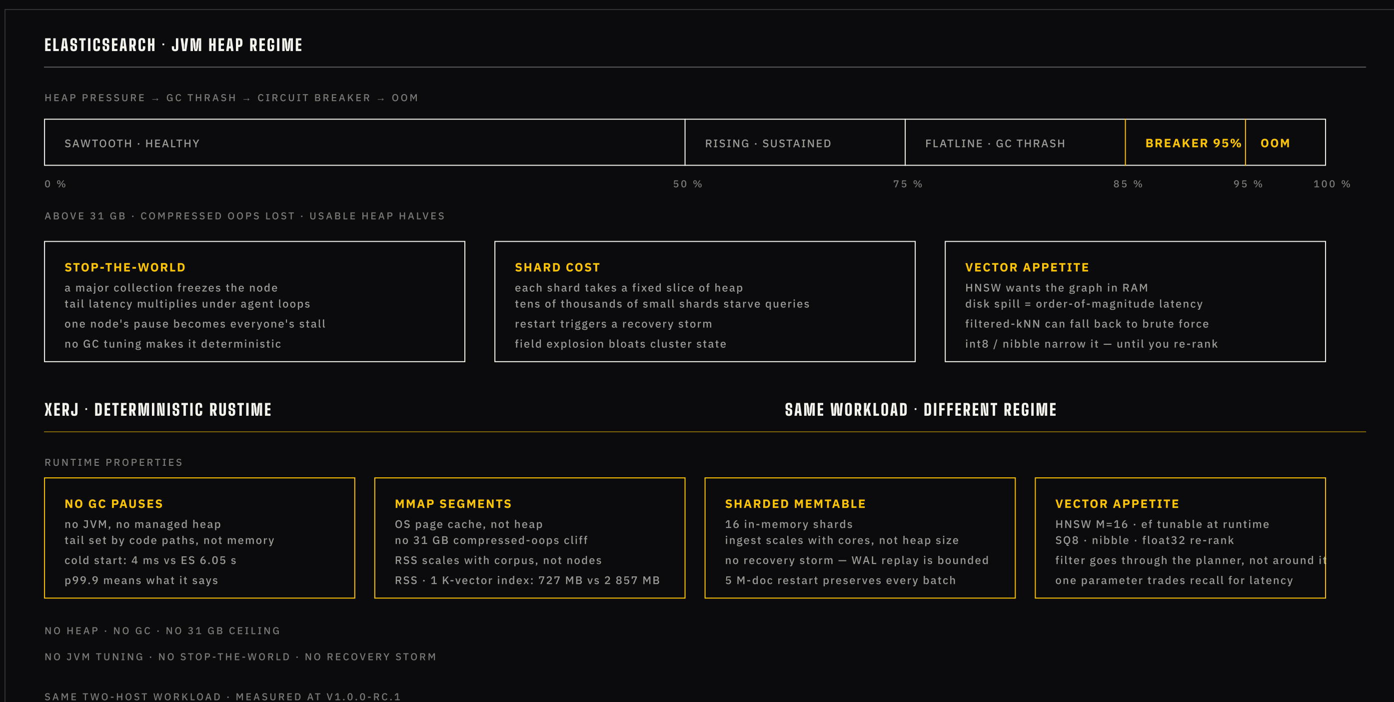
— DATA PATH — XERJ PATH · :9200 --- BOUNDARY

FROM THE CLIENT'S POINT OF VIEW · NOTHING CHANGES — THE NUMBER OF VENDORS DOES

04 · HEAP, GC, AND THE 31 GB WALL

A CEILING YOU CAN'T UPGRADE OUT OF.

The single most persistent source of instability in production Elasticsearch clusters is JVM memory management. Vector workloads raise the stakes: the HNSW graph wants the full vector set in RAM, filtered kNN can fall back to brute force, and a stop-the-world pause becomes a user-visible stall in every agent turn that touches the node.



— ELASTICSEARCH HEAP REGIME — XERJ RUNTIME --- ZONE BOUNDARY

SOURCE · ELASTIC TROUBLESHOOTING DOCS · SIRIUS OPEN SOURCE OPS NOTES · 2024–26

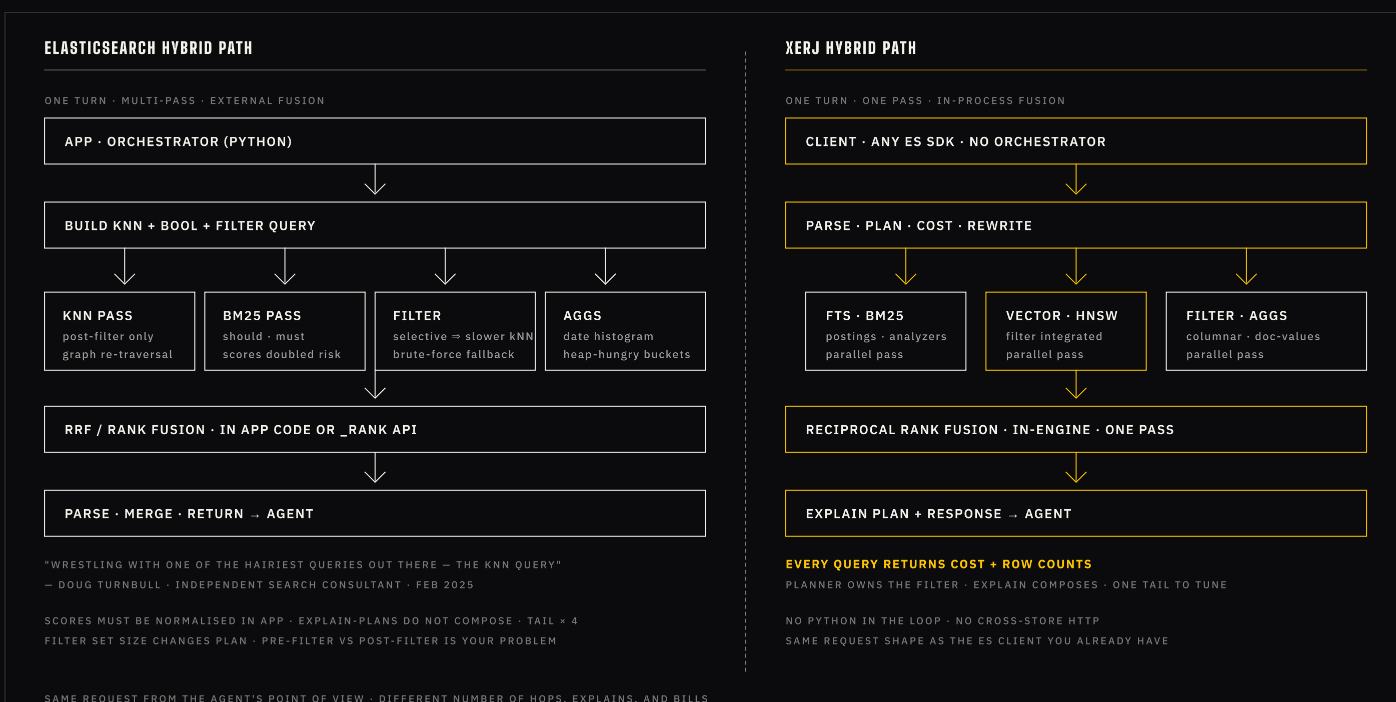
05 · HYBRID QUERY REALITY

FOUR HOPS.
OR ONE PLAN.

Hybrid retrieval — keyword + dense vector + filter + aggregation, fused — is the default request shape for an AI agent. In Elasticsearch it is several requests stitched together, each with its own scoring regime. In Xerj it is one request the planner decomposes, runs in parallel, and fuses in-engine.

BUILDING HYBRID SEARCH WITH ELASTICSEARCH REQUIRES WRESTLING WITH ONE OF THE HAIRIEST QUERIES OUT THERE — THE KNN QUERY.

DOUG TURNBULL · INDEPENDENT SEARCH CONSULTANT · SOFTWAREDOUG LLC · FEB 2025



— ES PATH   — XERJ PATH   --- BOUNDARY

06 · WHEN THE FILTER MAKES IT SLOWER

SEMI-RESTRICTIVE FILTERS BREAK HNSW.

Conventional databases get faster as filters get more restrictive. Filtered approximate kNN does the opposite: an HNSW graph traversal that has to skip non-matching nodes does *more* vector comparisons, not fewer, and can fall back to a brute-force scan. Architectures that depend on selective filters discover this the hard way — in production, on real corpora.

THE PERFORMANCE SIGNIFICANTLY DEGRADES, AND IN SOME CASES I ENCOUNTER RESPONSE TIMEOUT ERRORS (> 60 SECONDS).

SALEH ABUALI · ELASTIC DISCUSS USER · 200 M-DOC INDEX · 6 NODES · 34 GB HEAP / 66 GB RAM EACH · APR 2025

WHY THIS HAPPENS — AND WHAT XERJ DOES INSTEAD

ELASTICSEARCH	XERJ
FILTER PLACEMENT Post-filter on a candidate set; selective filters force re-traversal or brute force fallback. Recent ACORN work helps but does not erase the cliff for older 8.x deployments.	**FILTER PLACEMENT** Filter is a planner input, not a post-condition. Range, term, and exists predicates compose with HNSW traversal in one pass.
RECALL DIAL ef-search and num_candidates are settable, but tuning per-query requires re-thinking the request from the app side.	**RECALL DIAL** Single runtime parameter trades recall for latency without rebuilding the index. Same artifact serves chat UI and offline batch.
MEMORY APPETITE Graph wants RAM. Off-heap mmap helps; if it spills to disk, latency degrades. JVM heap and OS cache trade against each other.	**MEMORY APPETITE** SQ8 quantization (4× compression) is the working representation. Optional float32 re-rank pass restores precision on the final top-K.

07 · FOOTPRINT & COST

A FREE LICENCE IS NOT A FREE STACK.

The dominant cost in an ELK deployment is not the licence — it is the hardware, the sustained heap, the duplicated retention layers, and the engineering cost of running a JVM platform at scale. Independent analyst figures put a small ELK environment past two million dollars over three years; a large one north of sixty. Xerj is one static binary on commodity NVMe.

ES BINARY IMAGE	XERJ BINARY	RSS · 1 K VECTORS	COLD START
~600 MB	**19.6 MB**	**3.9×**	**863×**
JRE + PLUGINS + TOOLING	SINGLE STATIC · NO JVM	ES 2 857 MB · XERJ 727 MB	ES 6.05 S · XERJ 4 MS

3-YEAR ELK STACK TCO — INDEPENDENT ANALYST RANGE

ENVIRONMENT	YEAR 1	YEAR 2	YEAR 3	3-YEAR TCO
Small ELK	$427 K	$723 K	$1.22 M	$2.37 M
Large ELK · 20 TB / day	$15.85 M	$21.13 M	$28.54 M	$65.62 M

Source: ChaosSearch, "ELK Stack Costs Add Up: Here's How to Switch" — independent TCO model summarising hardware, retention, and operational labour. Elastic announced a pricing change effective 2025-01-27 with an estimated ~30 % increase for a typical production workload (Quesma analysis, 2025).

WHAT THE STATIC BINARY BUYS ·

NO JRE · NO PLUGIN MATRIX · NO LANGUAGE RUNTIME PAGER ROTATION

08 · VOICES FROM PRODUCTION · 2023–26

PUBLIC. DATED. UNEDITED.

A sample of public, attributed observations from architects, consultants, and forum operators running Elasticsearch at scale against AI workloads. Each is reproduced verbatim. Sources are listed on the closing page.

"It's a big ugly query with lots of repetition." DOUG TURNBULL · INDEPENDENT SEARCH CONSULTANT ON IMPLEMENTING HYBRID SEARCH IN ES · FEB 2025	*"It eats up alllll the resources (maxed out CPU and ate up loads of RAM)."* NDTREVIV · ELASTIC DISCUSS USER ES VS MILVUS AS A VECTOR DB · 2023
"A complex query might pass the pre-flight check but expand exponentially during execution, causing heap exhaust." SIRIUS OPEN SOURCE PROBLEMS & OPERATIONAL WEAKNESSES OF ELASTICSEARCH	*"If the JVM executes a major collection, the node is effectively frozen during a stop-the-world pause."* SIRIUS OPEN SOURCE ON GC BEHAVIOUR IN PRODUCTION CLUSTERS
"Accumulating tens of thousands of small shards is problematic because each shard consumes a non-trivial fixed cost of JVM heap." SIRIUS OPEN SOURCE ON SHARD ECONOMICS	*"Elasticsearch is great for traditional keyword search, but it struggles with semantic understanding [...] and latency at scale."* NIC SCHELTEMA · SHAPED.AI THE 7 BEST ELASTICSEARCH ALTERNATIVES IN 2025

EVERY QUOTE · PUBLIC URL · ATTRIBUTABLE AUTHOR · ORIGINAL CONTEXT — SEE PAGE 12

09 · NUMBERS YOU CAN REPRODUCE

THIRD-PARTY. SAME HARDWARE.

Two reference points. Trail of Bits, an independent security and systems firm, published a March 2025 benchmark comparing Elasticsearch 8.15.4 to OpenSearch 2.17.1 on the canonical Big5 query set. Xerj's v1.0.0-rc.1 verification run measured the same Elasticsearch build on the same NVMe host, using identical wire payloads — not extrapolated.

TRAIL OF BITS — ES 8.15.4 VS OS 2.17.1 — MARCH 2025

DIMENSION	ELASTICSEARCH	OPENSEARCH	DELTA	SOURCE
Big5 · geomean	baseline	1.6× faster	+60 %	trail of bits
Date histogram	baseline	16.55× faster	16.55×	trail of bits
Term aggregation	baseline	3.38× faster	3.38×	trail of bits
Vector search · default engine	baseline	11 % faster	+11 %	trail of bits

XERJ V1.0.0-RC.1 VS ELASTICSEARCH 8.13 — VERIFY RUN

DIMENSION	ES 8.13	XERJ	DELTA	SOURCE
Cold start	6.05 s	4 ms	863×	verify run
RSS · 1 K-vector index	2 857 MB	727 MB	3.9×	verify run
Index creation	93.9 ms	2.4 ms	39×	verify run
Top-K sort · p50	4.28 ms	0.34 ms	12.6×	verify run
Bulk · 1 K vectors	10 219 d/s	22 974 d/s	2.25×	verify run

EVERY ROW · REPRODUCIBLE WITH PUBLISHED SCRIPTS · SAME HOST · SAME WIRE

10 · VECTOR + FTS + HISTOGRAM + MEMORY

SAME WORKLOAD.
SAME ENGINE.

The four modalities an AI agent uses in a single turn — full-text ranking, dense-vector recall, time-series histograms, and agent memory — all live in Xerj behind one wire protocol, one segment cache, one permission model, one explain-plan. Elasticsearch covers the first three with mature implementations and a platform-grown ecosystem; agent memory remains an assemble-it-yourself problem.

CAPABILITY	ELK / ES + VECTOR DB	XERJ	NOTE
FULL-TEXT · BM25	FULL	FULL	38 ES query types · 10 analyzers
DENSE-VECTOR · HNSW	FULL native in 8.x · BBQ in 9.x	FULL SQ8 · nibble · re-rank · per-index metric	filter is a planner input
HYBRID · RRF FUSION	PARTIAL requires _rank or app-side merge	FULL in-engine, single pass, k=60 default	no Python orchestrator
DATE HISTOGRAMS · AGGS	FULL heap-bound · bucket explosions	FULL columnar · delta-of-delta timestamps	trail of bits: ES 16.55× slower
AGENT MEMORY · SEMANTIC DEDUP	NOT NATIVE redis + postgres + app code	FULL append-only · recency-blended scoring	one permission · one audit
EXPLAIN PLAN · COST · ROWS	PARTIAL does not compose across hybrid passes	FULL every query returns cost + counts	composes through fusion pass
RUNTIME · MEMORY MODEL	JVM HEAP 31 GB ceiling · GC pauses · circuit breakers	RUST · MMAP no heap · no GC · OS page cache	cold start: 4 ms vs 6.05 s ES
CONSENSUS · CO-PROCESS	EMBEDDED master nodes · cluster state can bloat	EMBEDDED RAFT metadata only · data plane stays out	no etcd / zk · either way
AUDIT · TAMPER-EVIDENT	PAID TIER audit logging in higher subscriptions	CORE SHA-256 hash chain · WORM · verify endpoint	core feature · in the binary
FOOTPRINT · INSTALL	~600 MB IMAGE JRE + plugins + tooling	19.6 MB STATIC single binary · air-gap · multi-arch image	one config · one metrics endpoint
WIRE COMPATIBILITY	CANONICAL	ES 8.13 DROP-IN 1 305 / 1 329 conformance cases	existing SDKs · Kibana connect

FULL · NATIVE, GA · PARTIAL · NATIVE BUT WITH ARCHITECTURAL CAVEAT · NOT NATIVE · REQUIRES EXTRA SYSTEM

11 · FROM CLUSTER TO BINARY

A CONFIG CHANGE. NOT A REWRITE.

Xerj speaks the Elasticsearch 8.13 wire protocol on port 9200 — bulk, scroll, kNN, hybrid, aggregations, the 38 query types your ingest collectors and dashboards already speak. The migration is structurally a swap of endpoint, not a rewrite of code.

A FOUR-WEEK PATH FOR THE TYPICAL AI WORKLOAD

WEEK 0 · SCOPING	Pick one workload — RAG, SIEM, observability, or agent memory. Confirm the wire surface in use (Bulk, kNN, RRF, aggs). Sign mutual NDA if required (not a precondition for the binary).
WEEK 1 · DEPLOY	Single binary or Helm chart in your environment. Reproduce the published numbers on sample data. Smoke-test the existing client SDKs (official Elasticsearch, LangChain, LlamaIndex). No code change required.
WEEK 2 · SHADOW	Mirror real traffic. Compare latency, footprint, and cost on identical hardware. Measure where the JVM tail used to live. Surface any compatibility gap as a triaged item, not a blocker.
WEEK 3 · DECISION	Decision meeting on your numbers. We hand over reproduction scripts, the Prometheus dashboards, and the audit-chain export so the result is independently verifiable inside your perimeter.

WHAT MIGRATION DOES NOT ASK

NOT REQUIRED	PROVIDED
CLIENT REWRITE Existing official Elasticsearch SDKs continue to work. LangChain and LlamaIndex retrievers connect unchanged. Kibana boards continue to query.	**DROP-IN ENDPOINT** :9200 with bulk, scroll, kNN, search, aggs. 38 query types. 1 305 / 1 329 conformance cases against the published ES 8.13 REST suite.
SHARD-COUNT TUNING No JVM heap tuning, no shard-count maths, no recovery-storm runbooks.	**ONE CONFIG FILE** One static binary. One metrics endpoint. One audit trail. One backup story. Liveness + readiness on both data ports.
NEW RUNBOOK Day-2 ops continue with the same primitives — Prometheus, Kubernetes probes, structured logs.	**SAME OPS SURFACE** Helm StatefulSet · PVC · headless service · multi-arch image · HEALTHCHECK wired to readiness · RBAC and audit included in the binary.

12 · GET STARTED · SOURCES

PROVE IT ON YOUR DATA.

We send the binary, the Helm chart, and the reproduction scripts for every number in this brief. You send one workload. We meet again in two weeks with your numbers, on your hardware, behind your perimeter.

CONTACT

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SOURCES — VERBATIM QUOTES & FIGURES IN THIS BRIEF

Doug Turnbull, "Elasticsearch hybrid search in practice" — `softwaredoug.com/blog/2025/02/08/elasticsearch-hybrid-search` — Feb 2025. Independent search consultant, SoftwareDoug LLC.
Saleh AbuAli, "Performance Issue with KNN + Filter on Large Index (v8.12)" — `discuss.elastic.co/t/377279` — Apr 2025. Production operator, 200 M-doc index.
Sirius Open Source, "Problems and Operational Weaknesses of Elasticsearch" — `siriusopensource.com/en-us/blog/problems-and-operational-weaknesses-elasticsearch`. JVM, sharding, recovery, and field-explosion observations.
Nic Scheltema, "The 7 Best Elasticsearch Alternatives in 2025" — `shaped.ai/blog/the-7-best-elasticsearch-alternatives-in-2025` — 2025.
ndtreviv, "Pros and Cons of using Elastic as a vector database" — `discuss.elastic.co/t/338733` — 2023. Forum operator comparing Elasticsearch to Milvus.
Evan Downing, Trail of Bits, "Benchmarking OpenSearch and Elasticsearch" — `blog.trailofbits.com/2025/03/06/benchmarking-opensearch-and-elasticsearch` — Mar 2025.
ChaosSearch, "ELK Stack Costs Add Up: Here's How to Switch" — `chaossearch.io/blog/switching-from-the-elk-stack-elasticsearch-costs`. 3-year ELK TCO model. Pricing change reference: Quesma analysis, 2025.
Elastic — official troubleshooting docs, "Managing and troubleshooting Elasticsearch memory" / "High JVM memory pressure" — `elastic.co/blog/managing-and-troubleshooting-elasticsearch-memory`. Heap, circuit breaker, and 31 GB compressed-OOPs ceiling.
Xerj — verify run, v1.0.0-rc.1. Single 32-core NVMe host, identical wire payloads, identical client. Reproduction scripts included with the pilot drop.